

İZMİR DEMOKRASİ UNIVERSITY



MIS203 / WEEK 01

First Programs And Problem Solving

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This Week In Python

- 1 Explain each line of a Python file
- 2 Use `print()`, variables and `input()`
- 3 Save and explain a first GitHub commit

A student introduction arrives as scattered answers. How can Python turn it into a clear card?

What would the program need to know?

A Program Follows Explicit Instructions

A useful program has a clear purpose.

1. Input: what information does it need?
2. Process: what should it do with that information?
3. Output: what should the user see?

Example: a student introduction card

First Lines Of Python

Example: examples/01_hello.py, then change one line.

```
print("Hello, MIS203!")  
print("A program runs instructions in order.")  
print(8 + 4)
```

Predict the three output lines first.

Variables Give Values Useful Names

A variable connects a name to a value.

1. Use names that describe the information.
2. Assignment uses =. Read it as “store this value”.
3. Reassign a name when a value changes.

Example: `examples/02_variables.py`

Input Is Text

`input()` pauses and waits for the user.

```
name = input("Your name: ")  
print(f"Welcome, {name}!")  
print(type(name))
```

What type does `input()` return?

Python Values Have Types

Text and numbers behave differently.

1. str stores text, including a typed student ID.
2. int and float store numbers.
3. bool stores True or False.

Compare 203 and "203" in examples/04_types.py

A Readable Student Card

An f-string places values into a sentence.

```
name = input("Name: ")
student_id = input("Student ID: ")
print(f"Student: {name}")
print(f"ID: {student_id}")
```

Add a department line.

Predict The Output

The quotation marks change what Python sees.

```
print(5 + 2)
print("5" + "2")
print("The answer is", 5 + 2)
```

Write all three output lines. Compare your prediction with the example.

Live Coding: Student Introduction

Build a program from a blank file.

1. Ask for name, student ID and one Python goal.
2. Print a clear card with all three answers.
3. Change one input and check the new output.

Compare with `project_a_student_profile.py` after you try.

Project A: Student Profile Builder

A complete terminal profile with a suggested repository name.

```
projects/project_a_student_profile.py
```

Extend it with one more field.

Project B: Meeting Brief

Turn raw meeting details into a useful agenda.

`projects/project_b_meeting_brief.py`

Change the output format for a second team.

In-Class Lab: Student Introduction Card

Your Challenge

Assignment: `week01/lab-quiz/lab_assignment.md`

Submit in your repository: `week01/lab-quiz/`

Build it. Check a boundary. Explain one decision.

GitHub Checkpoint

Lab and quiz: week01/lab-quiz/

Keep weekly homework in week01/ outside that folder.

```
git add week01/lab-quiz/ README.md  
git commit -m "week01: complete lab quiz"  
git push
```

Show the lab, one test and the commit.

See You Next Week